



Epic Calculator - User Documentation

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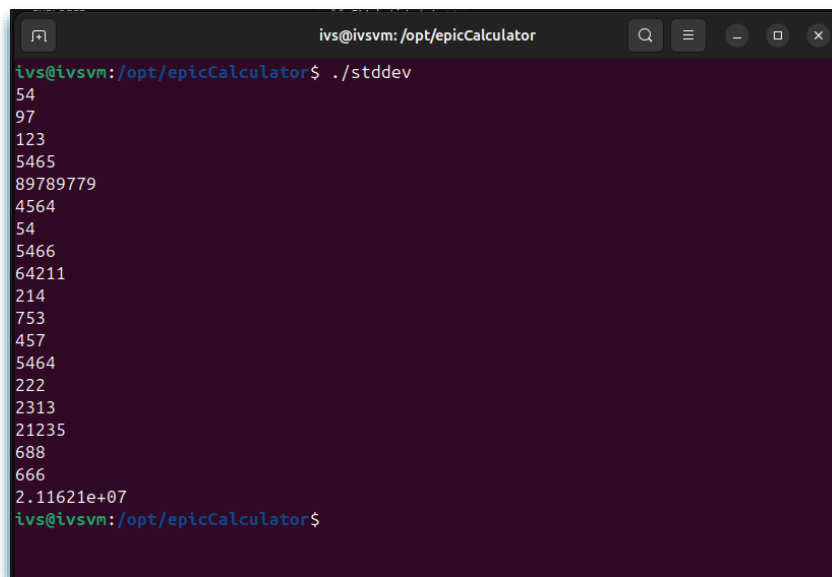
Product Overview

Epic Calculator offers a user-friendly interface and supports basic arithmetic operations (addition +, subtraction −, multiplication ×, division ÷), a factorial function, exponentiation with natural-number exponents, square roots, absolute value, and the modulo function. It also evaluates mathematical expressions with parentheses, following the correct order of operations.



Epic Calculator User Interface

Users can also install a command-line program called **stddev**, which calculates the standard deviation of the numbers provided as input.



Stddev program preview

Hardware and Software Requirements

Application is available only on Linux, the preferred distribution is Ubuntu 24.04 – 64bit.

The size of Epic Calculator after unpacking is approximately 213 KB.

The size of stddev after unpacking is approximately 76 KB.

For the Epic Calculator to run correctly, you must have Qt library version 6.2 or later installed on your device.

Installation and Uninstallation

Via installer

Installation

Epic Calculator

1. Download “epicCalculator-1.0.0-Linux-EpicCalc.deb”.
2. Open a terminal and navigate to your download directory.
3. Insert and run command: “sudo apt --fix-broken install ./epicCalculator-1.0.0-Linux-EpicCalc.deb”.
4. Follow the instructions on screen.

Epic Calculator will be installed to “/opt/epicCalculator”.

Launch it with: “/opt/epicCalculator/epicCalculator” or find “Epic Calculator” in your system menu.

Stddev

1. Download epicCalculator-1.0.0-Linux-stddev.deb
2. Open a terminal and navigate to your download directory.
3. Insert and run command: sudo apt --fix-broken install ./epicCalculator-1.0.0-Linux-stddev.deb”.
4. Follow the instructions on screen

stddev will be installed to “/opt/epicCalculator”.

Launch it with: “/opt/epicCalculator/stddev” .

Uninstallation

1. Open a terminal
2. Insert run command “sudo apt remove epiccalculator-epiccalc” or “sudo apt remove epiccalculator-stddev” depending which application you want to remove.
3. Insert your password.
4. Type “Y” to confirm removal.

Building and Installing from Source

First of all, clone project repository to one of your directories with command: “git clone <https://github.com/Ekuuri/ivs-proj2.git>” or download and unpack .zip file of the project
Then follow these steps:

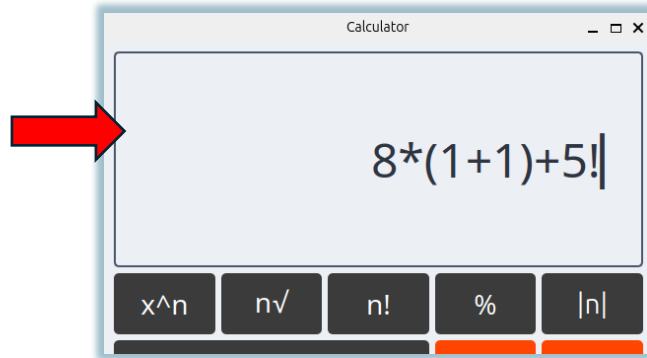
1. Enter the source directory with “cd src”
2. Build all targets with “make all”. If the build fails, follow the instructions shown in the terminal.
3. Install the component you want to:
 - a. Just Epic Calculator: “cmake --install ../build --component EpicCalc”
 - b. Just stddev program: ”cmake --install ../build --component stddev”
 - c. Both : ”cmake --install ../build”

Product Usage

Epic Calculator

You can launch the application from your system menu (look for “Epic Calculator”) or by running: “/opt/epicCalculator/epicCalculator“

Once launched, the main interface will appear. Your input and results are shown on the main display.



Calculator display

You can enter expressions via the on-screen buttons or your keyboard. See the table below for a list of buttons, their functions, and corresponding keyboard shortcuts.

Buttons	Display Output	Function	Keybinds
0-9	Corresponding number	—	Number keys
+	+	Adds numbers	+
-	-	Subtracts numbers	-
×	*	Multiplies numbers	*
÷	/	Divides numbers	/
.	.	Decimal point	.
(	(	Opens a bracket for expression grouping	(
)	)	Closes bracket	)
←	(Moves left)	Moves cursor one position left	Left arrow
→	(Moves right)	Moves cursor one position right	Right arrow
=	(Evaluates)	Calculates the result	=
x^n	^	Raises a number to n'th power	^
n√	√	N'th root	(Unbound)
n!	!	Factorial of a number	!
%	%	Division remainder (modulo)	%
n		Absolute value (needs to be in pair)	
⌫	(Deletes)	Deletes last character	Backspace
C	(Clears input)	Clears full input	Esc
CE	(clears input)	Clears input after last operator (+ - * /)	Esc
?	(Usage help)	Showes help message	?

Buttons explanation table

In case of any error, error message will show up on the display of calculator. A more detailed description of each error message is in an information table below.

Error Message	Error Description
Error: Division by zero	A division operation attempted to divide by zero, which is undefined.
Error: Factorial of a negative number	Factorials are only defined for non-negative numbers.
Error: Factorial of a non-integer number	Factorials must be applied to whole numbers only.
Error: Not a natural exponent	Exponentiation requires a natural (non-negative) exponent for certain base values.
Error: Not a real number	The result is not a real number (e.g., even root of a negative number).
Error: Cannot compute 0th root	Zero-th root is undefined in mathematics.
Error: Invalid input	The entered expression is not valid or incomplete.
Error: Mismatched parentheses	Parentheses were opened or closed incorrectly.
Error: Mismatched absolute value bars	Absolute value bars () are unbalanced or misplaced.

Error messages table

Built-in help is available by clicking the “?” icon in the interface.



Built-in help

About the Authors

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